MULTI-AGENT DATA ANALYSIS AND INSIGHT GENERATION PLATFORM FOR FINANCIAL AND SALES DATA

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Presentation Outline

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- LLM Architecture
- Methodology
- Current Progress
- Mid-Progress Timeline
- Remaining Work
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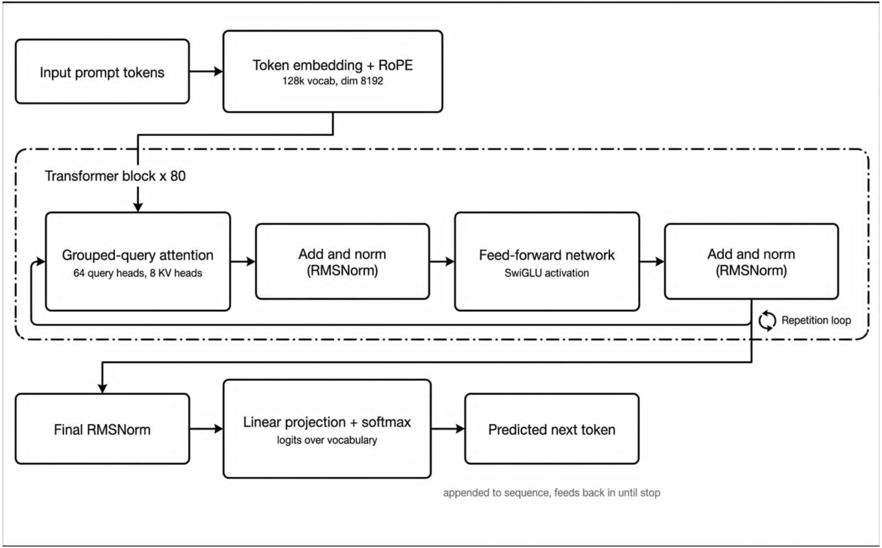
Introduction

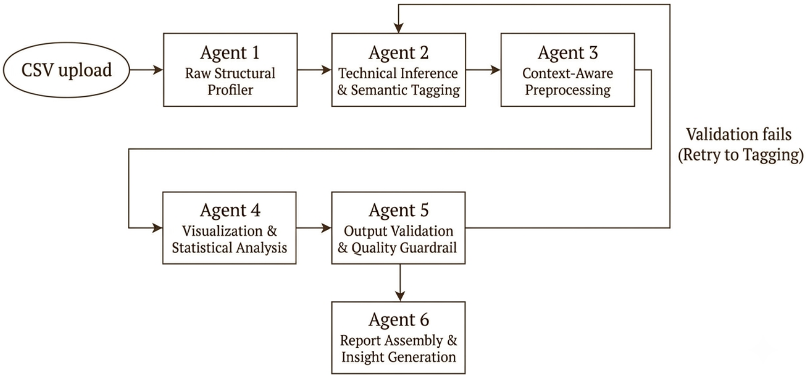
- The project is being implemented as a state-based data analysis pipeline for sales and financial CSV datasets.
- Four pipeline stages are currently connected: structural profiling, semantic tagging, preprocessing, and statistical analysis.
- The system transforms raw records into a cleaned dataset, a preprocessing audit trail, quality metrics, statistics, and chart files.
- Mathematical operations are performed with Python libraries; the language model is used for semantic interpretation and processing rules.
- The current implementation provides a working backend foundation for the planned user-facing platform.

LLM Architecture – Llama Transformer

- The model used in the pipeline is Llama, a decoder-only transformer language model.
- It processes the input context through repeated self-attention and feed-forward blocks.
- The attention layers help the model relate each token to earlier tokens in the prompt.
- This architecture makes it suitable for semantic tagging, schema inference, and structured reasoning.


Input prompt tokens
Token embedding + RoPE
128k vocab, dim 8192
Transformer block x 80
Grouped-query attention
64 query heads, 8 KV heads
Add and norm
(RMSNorm)
Feed-forward network
SwiGLU activation
Add and norm
(RMSNorm)
Repetition loop
Final RMSNorm
Linear projection + softmax
logits over vocabulary
Predicted next token
appended to sequence, feeds back in until stop

Methodology – Current 4-Stage Pipeline



Current Progress

Implementation of the 4-Stage Pipeline

- Agent 1: Raw Structural Profiler
- Agent 2: Technical Inference & Semantic Tagging
- Agent 3: Context-Aware Preprocessing
- Agent 4: Visualization & Statistical Analysis

Agent 1: Raw Structural Profiler

- Reads the input CSV and records shape, dtypes, missingness, uniqueness, and duplicates.
- Produces a raw structural profile without changing the source data.
- Caches the DataFrame for the downstream stages.

```
AGENT 1 — Structural Profile

Rows:             10000
Columns:          21
Total cells:      210000
Missing rate:     0.0%
Duplicate rows:   0 (0.0%)
Column profiles: 21 columns profiled
```

```
AGENT 1 — Structural Profile

Rows:             1067371
Columns:          9
Total cells:      9606339
Missing rate:     2.58%
Duplicate rows:   12133 (1.14%)
Column profiles: 9 columns profiled
```

Agent 2: Technical Inference & Semantic Tagging

- Sniffs numeric, datetime, boolean, and string types using local heuristics.
- Uses Groq-based semantic tagging when available and falls back to deterministic rules when it is not.
- Creates a per-column blueprint containing type, semantic tag, scaling, imputation, and analysis policies.
- Passes the blueprint to preprocessing instead of hard-coding one rule for every column.

```
AGENT 2 — Schema Blueprint

identifier                3 columns
unknown                   2 columns
text                      1 columns
count                     1 columns
datetime                  1 columns
currency                  1 columns
geographic                1 columns
Total tagged:    9 columns
```

```
┌ Country
  semantic_tag     : geographic
  intended_type    : string
  imputation       : none
  imputation_reason: no missing values, preserve for review
  encoding         : none
  identifier       : False
  analysis_allowed : True
  confidence       : 40.0
  null_action      : flag_only
  suitable         : True
└ reason           : text_or_label
```

Agent 3: Context-Aware Preprocessing

- Coerces columns according to the semantic blueprint and cleans international currency formats.
- Standardizes text and null variants while preserving identifiers.
- Removes duplicates, applies configured imputation, and clips permitted outliers with IQR bounds.
- Extracts date features and derives metrics such as profit, margin, and revenue per unit.
- Records scaling parameters, an audit log, and a 0–100 data quality score.

```
AGENT 3 — Preprocessing

Quality score:   98.47
Completeness:    None%
Duplicates removed: None
Cleaned shape:   812368 rows × 21 cols
Preprocessing steps logged: 21
Cleaned CSV:     outputs\cleaned_data.csv
```

```
[Agent 3] Preprocessing: profile=strict domain=finance_sales input=1067371x9
[Agent 3] Completed: rows=1067371->812368 cols=9->21 quality=98.47/100 raw_missing=2.58% remaining_nulls
=0.0%
[Agent 3] Cleaning summary: deduped=12133, scaled=0, added_cols=12, removed_rows=255003
[Agent 3] Note: the cleaned dataset is normalized and exported to outputs/cleaned_data.csv; if the row c
ount stays the same, that means there were no exact duplicates or rows to drop.
```

Agent 4: Visualization & Statistical Analysis

- Calculates descriptive statistics, correlations, distributions, growth, and seasonality.
- Detects anomalies using a configurable z-score threshold.
- Computes regression trends and category rankings when the dataset supports them.
- Saves reproducible PNG charts and structured statistics to the pipeline state.

```
[Agent 4] Starting analysis: 812368 rows × 21 cols
[Agent 4] Step 1 — Descriptive stats: 10 columns
[Agent 4] Step 2 — Correlation done, strong pairs: 0
[Agent 4] Step 3 — Growth rates skipped (no revenue/time axis)
[Agent 4] Step 4 — Rankings skipped (no revenue/category pairing)
[Agent 4] Step 5 — Seasonality skipped (no time axis)
[Agent 4] Step 6 — Anomalies: 23915 unique rows (2.94%) across 3 column
[Agent 4] Step 7 — Distributions done (0 charts)
[Agent 4] Step 8 — Regression skipped (no time axis)
[Agent 4] Step 9 — Distribution charts done (1 charts)
[Agent 4] Step 10 — Derived metrics skipped (no derived metrics)
[Agent 4] Done — 1 charts saved to outputs/charts/
```

```
AGENT 4 — Statistical Analysis

Descriptive stats columns: 10
Strong correlation pairs:  0
Anomalous rows:  23915 (2.94%)
Charts saved:    1
  outputs/charts\boxplot_numeric_cols.png
```

Mid-Progress Timeline

- **Completed:** project architecture, GraphState, Agents 1–3, conditional pipeline routing, preprocessing audit trail, quality scoring, and focused tests.
- **Current:** Agent 4 statistical analysis and chart generation, verified on the Amazon sales dataset.
- **Next:** connect statistics to evidence-backed insight generation.
- **Final phase:** upload workflow, frontend presentation, broader datasets, and end-to-end evaluation.

Remaining Work

Work package	Target outcome
Validation guardrail	Cross-check statistics, charts, and evidence
Insight generation	Produce traceable natural-language findings
Interface	Upload data and display progress/results
Evaluation	Benchmark accuracy, robustness, and runtime

Conclusion

- **Backend foundation achieved:** A working LangGraph pipeline now carries a CSV dataset through four specialized stages.
- **Core transformation achieved:** Profiling, semantic tagging, context-aware preprocessing, quality scoring, and statistical chart generation are operational.
- **Evidence available:** The verified run produces cleaned data, diagnostics, audit records, statistics, and chart artifacts.
- **Next objective:** complete validation, evidence-backed report generation, interface integration, and broader evaluation.

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